HACKTN26 · CHALLENGE #1 · SURPLUS TO SOCIAL GOOD

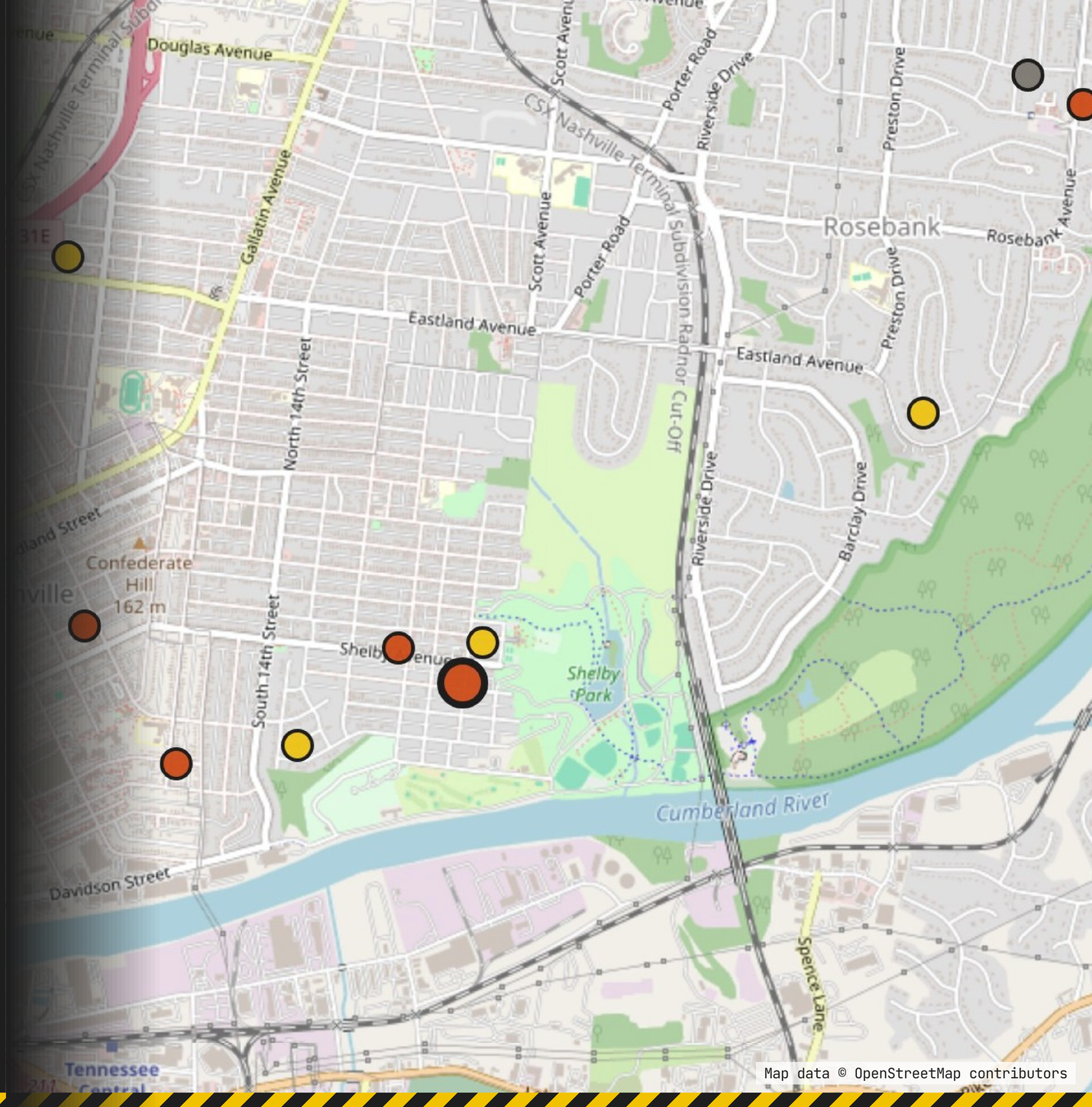
BEFORE THE BULLDOZER

Every dot on this map is a Nashville house being torn down. **We look inside before the bulldozer does.**

432,065
TONS OF C&D DEBRIS · NASHVILLE 2021

~3%
RECYCLED

Team West End Demons · beforethebulldozer.duckdns.org



THE PROBLEM

NASHVILLE BURIES ITS HOUSES

Construction and demolition debris is a fast-growing waste stream, and almost none of it is reused.

1 in 3

tons Nashville landfills is **construction & demolition debris**. The stream doubled in a decade.

97%

of C&D debris is **landfilled**: 13,928 of 432,065 tons were recycled in 2021.

90%

went to the Southern Services landfill until 2022. It borders **Bordeaux**, a predominantly Black neighborhood that has fought it for 20 years.

5–10

years of life left at **Middle Point**, the landfill that takes Nashville's trash.

THE GAP: THE TEARDOWN BOOM IS EXEMPT

Metro's proposed C&D ordinance excludes single-family homes and townhomes of 4 units or fewer. Nobody inventories the house before it comes down, so doors, cabinets, floors and fixtures go straight into the dumpster.

647

residential demolition permits issued in Nashville in the last 12 months (Metro open data)

Sources: Metro Nashville · Nashville Scene · WPLN · Waste Advantage · Metro Codes Building Permits Issued

PROBLEM VALIDATION

FILL IN REAL QUOTES BEFORE PRESENTING

WHAT WE HEARD

We asked: what happens to the stuff inside a teardown? What stops anyone from saving it? Would a salvage list plus a nonprofit pickup change that?

HABITAT FOR HUMANITY RESTORE

“Add the exact quote from the call here.”

WHAT IT CHANGED

e.g. which categories they accept, pickup rules

NASHVILLE DEMOLITION CONTRACTOR

“Add the exact quote from the call here.”

WHAT IT CHANGED

e.g. tipping fees, timeline, liability

REALTOR / HOMEOWNER / MENTOR

“Add the exact quote from the call here.”

WHAT IT CHANGED

e.g. who would use it, what would make them say no

Roles only, no full names · conversations held Oct 17–18, 2026

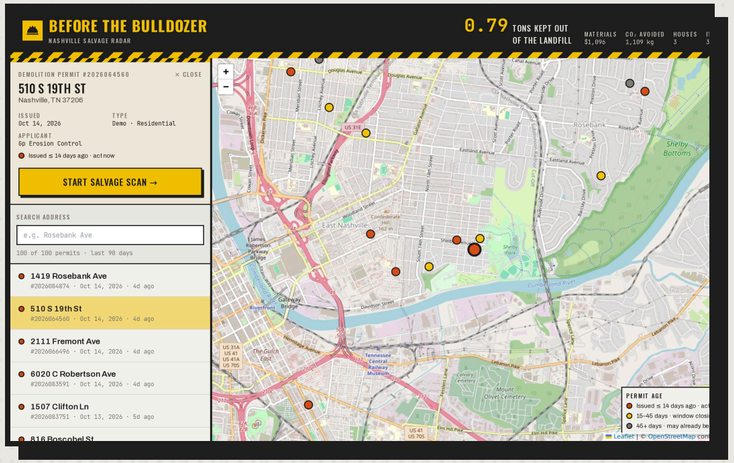
LIVE DEMO

THREE SCREENS. UNDER 20 SECONDS.

01

SALVAGE RADAR

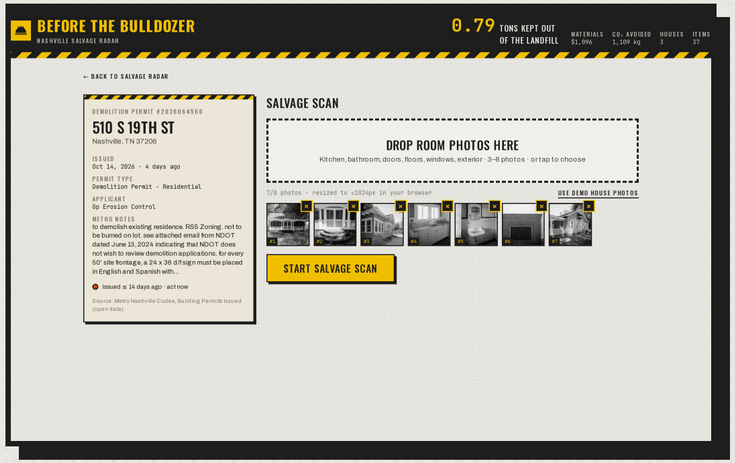
Real Nashville demolition permits from Metro open data, colored by how recently they were issued.



02

SNAP THE ROOMS

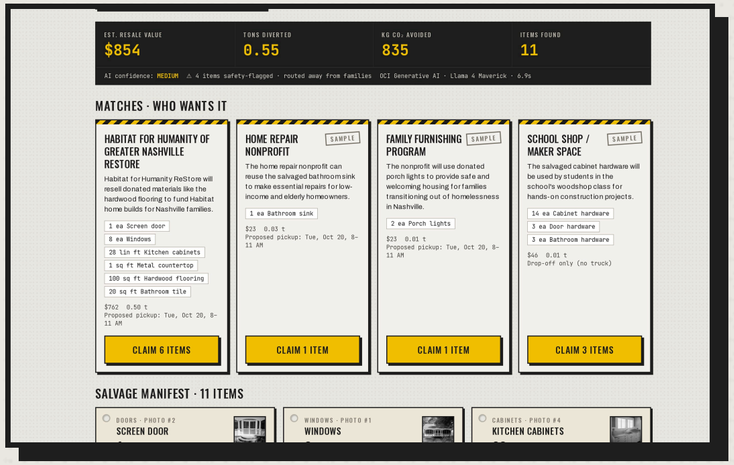
Kitchen, bath, doors, windows, exterior. Photos shrink in the browser before upload.



03

MANIFEST + MATCH

Items, condition, safety flags, value and tons, each routed to one nonprofit. Claim, and the citywide counter ticks up.



Demo house: 1920s bungalow, Historic American Buildings Survey photos (public domain)

TRY IT LIVE

beforethebulldozer.duckdns.org

HOW IT WORKS · BUILT ON ORACLE CLOUD

THE AI LOOKS. OUR RULES DECIDE.

METRO OPEN DATA
Residential demolition permits, ingested by script

→

NEXT.JS ON OCI COMPUTE
API routes · zod validation
JSON retry · 30 s fallback
rules engine · lookup table

⇄

OCI GENERATIVE AI
Llama 4 Maverick (vision)
photos → salvage manifest
us-chicago-1 · on-demand

→

NONPROFITS
Habitat ReStore + partners
claim → pickup window

AI ONLY COUNTS

Llama 4 identifies what's visible: item, quantity, condition, safety flags. It never invents a price.

OUR TABLE DOES THE MATH

Every number is traceable to a per-category lookup.

```
value = $/unit × qty × condition
weight = lb/unit × qty
```

RULES ROUTE SAFELY

Each item goes to exactly one nonprofit. Possible lead paint or asbestos never goes to families; a licensed pro assesses it.

6–7 s	1 + N	≤ 25	100
PER 7-PHOTO SCAN, END TO END	VISION CALL + ONE SHORT TEXT CALL PER MATCH	ITEMS PER MANIFEST, ZOD-VALIDATED	REAL PERMITS ON THE MAP

OCI Compute · OCI Generative AI · Autonomous Database as the production data store

IMPACT + WHAT'S NEXT

ONE HOUSE, THEN THE WHOLE CITY

ONE HOUSE · 7 PHOTOS

$854 REUSABLE MATERIALS

0.55 TONS DIVERTED

11 items routed to 4 nonprofits. Conservative: only what is visible in seven photos.

×647

NASHVILLE · ONE YEAR OF PERMITS

~$550K OF MATERIALS

~356 TONS KEPT OUT

Illustrative: 647 residential demolition permits (last 12 months) × the demo house.

PILOT

Run real teardowns with Habitat ReStore; confirm categories and prices.

AUTO-ALERTS

Every new demolition permit opens a salvage opportunity automatically.

ON-SITE CHECK

A human confirms the AI inventory before any truck is sent.

LIABILITY

Waiver template and lead/asbestos referral to licensed pros.

LOOK INSIDE BEFORE THE BULLDOZER DOES.

QUESTIONS

ASK US ANYTHING

HOW ACCURATE IS A PHOTO ESTIMATE?

It's a starting inventory a human confirms on site. We show AI confidence and never auto-dispatch a truck.

WHY WOULD A CONTRACTOR BOTHER?

Tipping fees: every ton not dumped is money saved, plus possible tax-deductible donation receipts.

WHERE'S ORACLE?

OCI Generative AI (Llama 4 Maverick) for vision and text, OCI Compute hosting, Autonomous Database as the production store.

WHAT ABOUT LEAD AND ASBESTOS?

Pre-1980 painted wood and old tile get flagged and routed away from families. Abatement needs a licensed pro.

HOW DOES IT SCALE?

Permits are already public. Any city with open permit data can turn each new demolition into a salvage opportunity.

WHAT'S REAL VS. SAMPLE?

Permits are real Metro data and Habitat ReStore is real. Three partner nonprofits are labeled samples; prices are estimates to verify.

LIVE APP
`beforethebulldozer.duckdns.org`

CODE
`github.com/mateenbajlory/before-the-bulldozer`

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